LECTURE 3a / Fall 2026

CHEME 5660
Cornell University
September 8, 2026
Jeffrey D. Varner

Equity Markets and the Stylized Facts of Growth Rates

LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Describe equity claims.** Explain what common stock represents and distinguish the corporation, an exchange, an index, and an exchange-traded fund.
- **Compute growth rates.** Calculate continuously compounded growth rates and convert them to log returns using $r = (\Delta t)g$.
- **Test return models.** Use return distributions and autocorrelation to check whether a model matches observed market data.

1 Why Corporations Sell Ownership Claims

Large ventures often need more capital and more time than one owner can provide. A corporation can raise that capital by dividing ownership into shares that investors can trade. A common shareholder may receive voting rights and dividends, but creditors are paid first if the corporation is liquidated. Limited liability usually limits the shareholder's loss to the amount invested; it does not prevent the share price from falling to zero.

An *equity security* is therefore an ownership claim on what remains after the corporation meets its contractual obligations. It is not a promise of fixed repayment. Its value depends on uncertain future distributions and the price another investor may later pay. The market price is the price available to buyers and sellers at a particular time and trading venue.

An exchange supplies listing and trading rules and an organized venue for matching orders. Clearing and settlement complete the obligations created by trades, often through specialized infrastructure distinct from the matching engine. A stock index, such as the S&P 500, is a rule-based numerical portfolio benchmark; it cannot itself be purchased. An exchange-traded fund (ETF) is a security whose shares trade intraday and whose portfolio may seek to track an index. These objects interact, but they are not interchangeable.

2 Growth Rates and Returns

A price change in dollars depends on the initial share price. We use the continuously compounded growth rate as the primary reward variable and reserve r for the dimensionless log return over a specified interval.

Definition 2.1. Growth rate and log return

Let $S_{t-1} > 0$ and $S_t > 0$ be consecutive share prices separated by $\Delta t > 0$ years. The continuously compounded growth rate g_t and corresponding log return r_t are

$$g_t := \frac{1}{\Delta t}\log\left(\frac{S_t}{S_{t-1}}\right), \qquad r_t := (\Delta t)g_t = \log\left(\frac{S_t}{S_{t-1}}\right). \tag{1}$$

Thus g_t has units inverse years and r_t is dimensionless. If dividends are included, use a consistently adjusted total-return price series before applying the definition.

The choice between price return and total return matters. A price-only series omits dividends. An adjusted-price series may include dividends and splits according to the data vendor's convention. Document those adjustments before comparing statistics.

Log returns add across nonoverlapping time intervals (Prop. 2.1); equivalently, cumulative log return is a time-weighted sum of growth rates.

Proposition 2.1. Aggregation of price log returns

Let $S_0, S_1, \ldots, S_T$ be strictly positive prices and let Δt_t be the length of interval t. Define $g_t := (1/\Delta t_t)\log(S_t/S_{t-1})$ and $r_t := (\Delta t_t)g_t$. Then

$$\sum_{t=1}^{T} r_t = \sum_{t=1}^{T}(\Delta t_t)g_t = \log\left(\frac{S_T}{S_0}\right), \tag{2}$$

which is the course convention $r = (\Delta t)g$ applied interval by interval.

Derivation. Use the logarithm of a product and cancel the intermediate price ratios:

$$\sum_{t=1}^{T}\log\left(\frac{S_t}{S_{t-1}}\right) = \log\left(\prod_{t=1}^{T}\frac{S_t}{S_{t-1}}\right) = \log\left(\frac{S_T}{S_0}\right).$$

The result is exact; the common approximation $\log(1+R_t) \approx R_t$ is only local for small $|R_t|$. □

An excess growth rate compares the equity with a benchmark rate over the same interval. If $g_{f,t}$ is the benchmark growth rate, then $\bar{g}_t := g_t - g_{f,t}$ and the corresponding excess log return is $\bar{r}_t = (\Delta t)\bar{g}_t$.

3 Common Patterns Across Many Assets

For m dates and n assets, collect centered returns in a matrix $G \in \mathbb{R}^{m\times n}$. Its singular value decomposition $G = U\Sigma V^{\top}$ writes the data as ordered rank-one patterns. Large singular values identify common patterns that account for much of the variation across assets. SVD finds patterns in the data; it does not explain their economic causes. Results depend on centering, scaling, missing-data treatment, and the selected assets.

COMPANION NOTEBOOK

L3a: SVD of S&P 500 Growth Rates →

Construct a multi-asset growth-rate matrix and use singular values and singular vectors to study common market patterns.

4 Reward, Risk, and the Measurement Horizon

The sample mean estimates average return under a chosen sampling rule; the sample standard deviation measures dispersion around that sample mean. Neither statistic alone describes downside magnitude, tail probability, drawdown, or liquidity. Moreover, annualizing daily volatility by multiplying by the square root of the number of periods relies on assumptions about scaling and dependence that empirical data may violate.

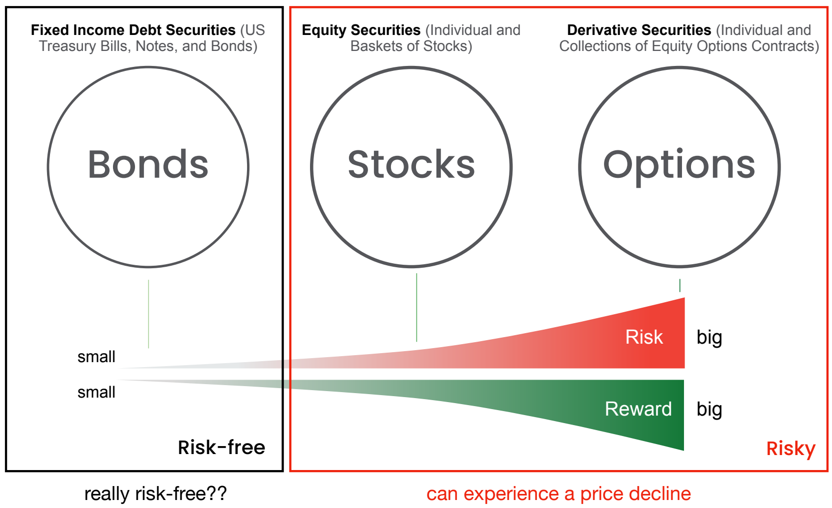

Figure 1: A conceptual risk–reward map. Expected reward and measured dispersion summarize different dimensions of an uncertain asset; assets with similar means and standard deviations can still have very different tail behavior, drawdowns, and liquidity.

Do not ask only "how volatile is this asset?" Ask what type of loss matters, over what horizon, and whether the chosen statistic will detect it. Value at risk, expected shortfall, maximum drawdown, downside deviation, beta, and tracking error answer different questions.

5 Stylized Facts as Model Requirements

A stylized fact is a recurring pattern observed across many assets or markets. It is not an exact law for every ticker or sample. A useful model should be able to reproduce the patterns that matter for the question being studied.

Heavy tails. Large-magnitude returns occur more frequently than a fitted Gaussian model predicts. Sample kurtosis, quantile comparisons, tail plots, and tests such as Anderson–Darling can reveal distributional mismatch, but tail-index estimation is sensitive to the threshold and sample size. "Heavy-tailed" should be supported by a stated mathematical or empirical criterion rather than used as a synonym for volatile.

Weak linear autocorrelation of raw returns. At ordinary liquid-market horizons, signed returns often show small linear autocorrelations. This does not establish independence: nonlinear dependence, market microstructure effects, and conditional predictability can remain. Statistical significance also depends on the sample length and simultaneous testing across lags.

Volatility clustering. Large absolute or squared returns tend to occur near other large movements, and calm observations near other calm observations. The signed return may have little autocorrelation while $|g_t|$ or g_t^2

remains strongly dependent. Volatility is therefore conditional and time-varying rather than a fixed property revealed by one unconditional standard deviation.

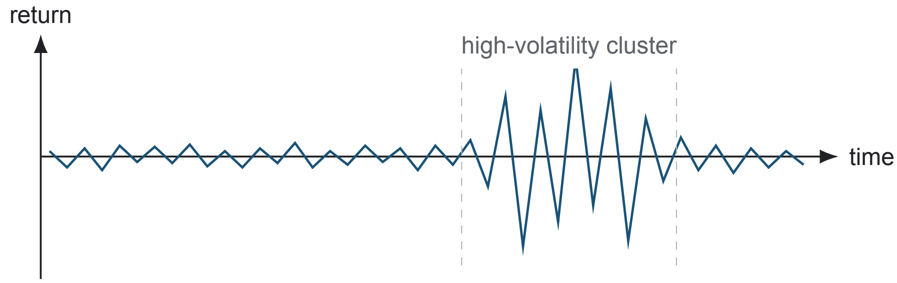

Figure 2: A schematic return path with changing conditional dispersion. The high-volatility interval contains both positive and negative observations, so clustering concerns return magnitude rather than a persistent return sign.

The empirical notebook calculates these diagnostics for common stocks and ETFs. Use the results to check the assumptions and predictions of the models introduced in L3b and later lectures.

COMPANION NOTEBOOK

L3a: Stylized Facts of Stocks and ETFs →

Compare Normal, Laplace, and Student-t fits; inspect Hill tail-index stability; and contrast signed-return with magnitude autocorrelation.

KEY TAKEAWAYS

In this lecture, we treated equity as a residual ownership claim, constructed consistent return measures, and used recurring empirical patterns to challenge models.

- **Returns require conventions.** Price versus total return, sampling interval, corporate-action adjustment, and compounding must be stated before statistics are compared.
- **Dependence hides beyond returns.** Small autocorrelation in signed returns does not imply independence; absolute and squared returns often reveal persistent conditional volatility.
- **Models should match observed patterns.** Heavy tails and volatility clustering reveal risks that a fixed-variance Gaussian or i.i.d. model can understate.

Next, build a binomial equity lattice and separate real-world probabilities used for forecasting from risk-neutral probabilities used for pricing.

ADDITIONAL READING

- Rama Cont, "Empirical Properties of Asset Returns: Stylized Facts and Statistical Issues," *Quantitative Finance* 1(2), 223–236 (2001).
- Benoit Mandelbrot, "The Variation of Certain Speculative Prices," *Journal of Business* 36(4), 394–419 (1963), for the early challenge to Gaussian return models.
- U.S. Securities and Exchange Commission, *Stocks*, for a concise description of shareholder claims and risks.